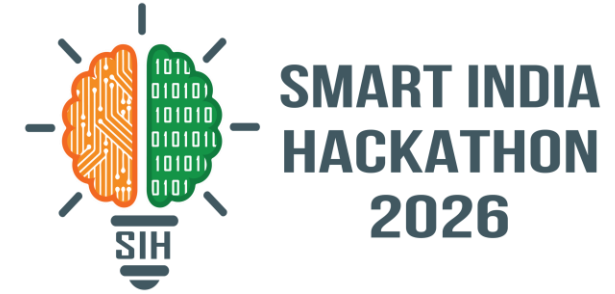


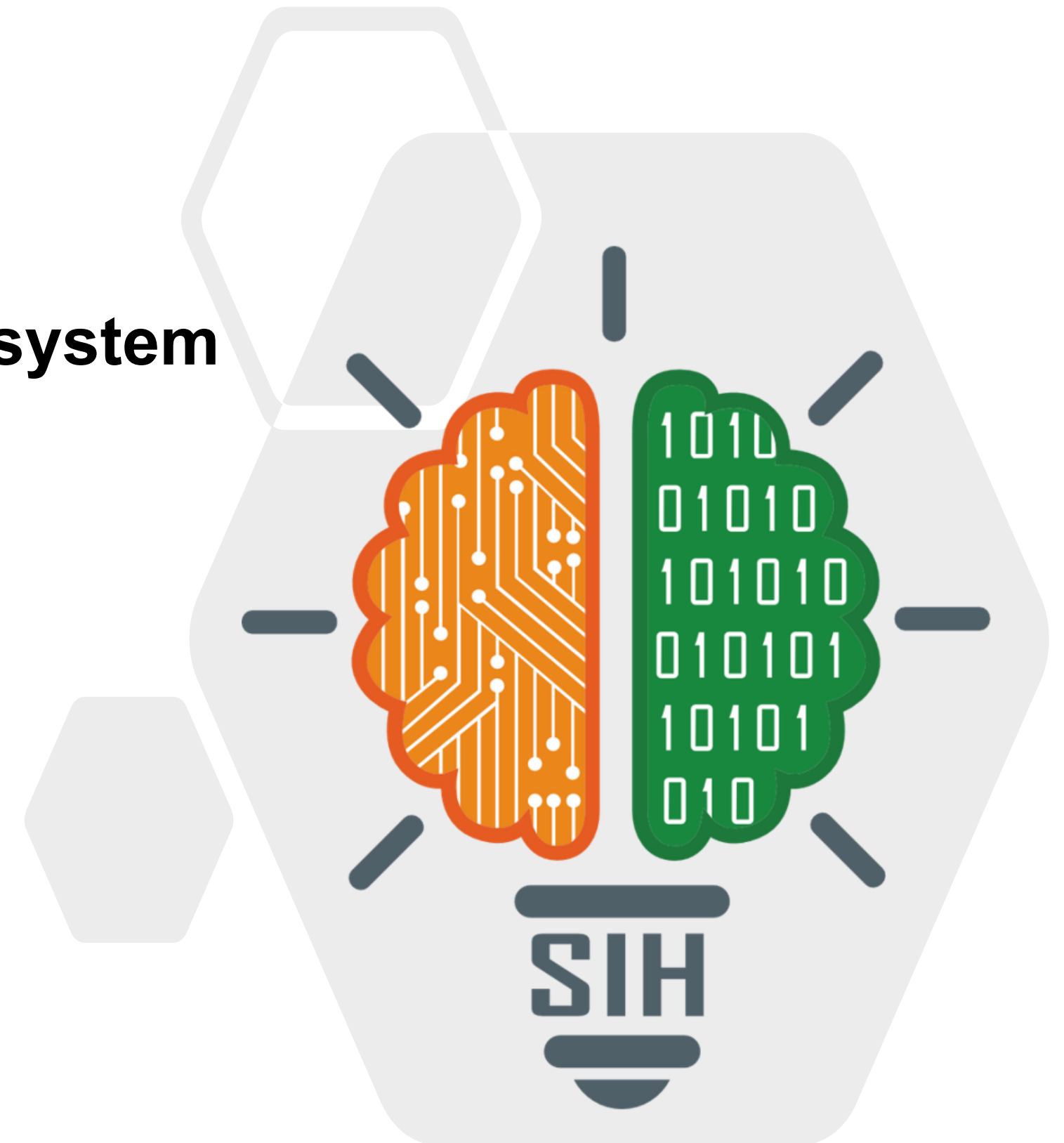
SMART INDIA HACKATHON 2026



- **Problem Statement ID - SIH26176**
- **Problem Statement Title - ORCA Marine EcOsysteem**

Reasoning with Collaborative Agents

- **Theme - Smart Automation**
- **PS Category - Software**
- **Team ID - 170524**
- **Team Name - Tech Udaan**



1 Proposed Solution

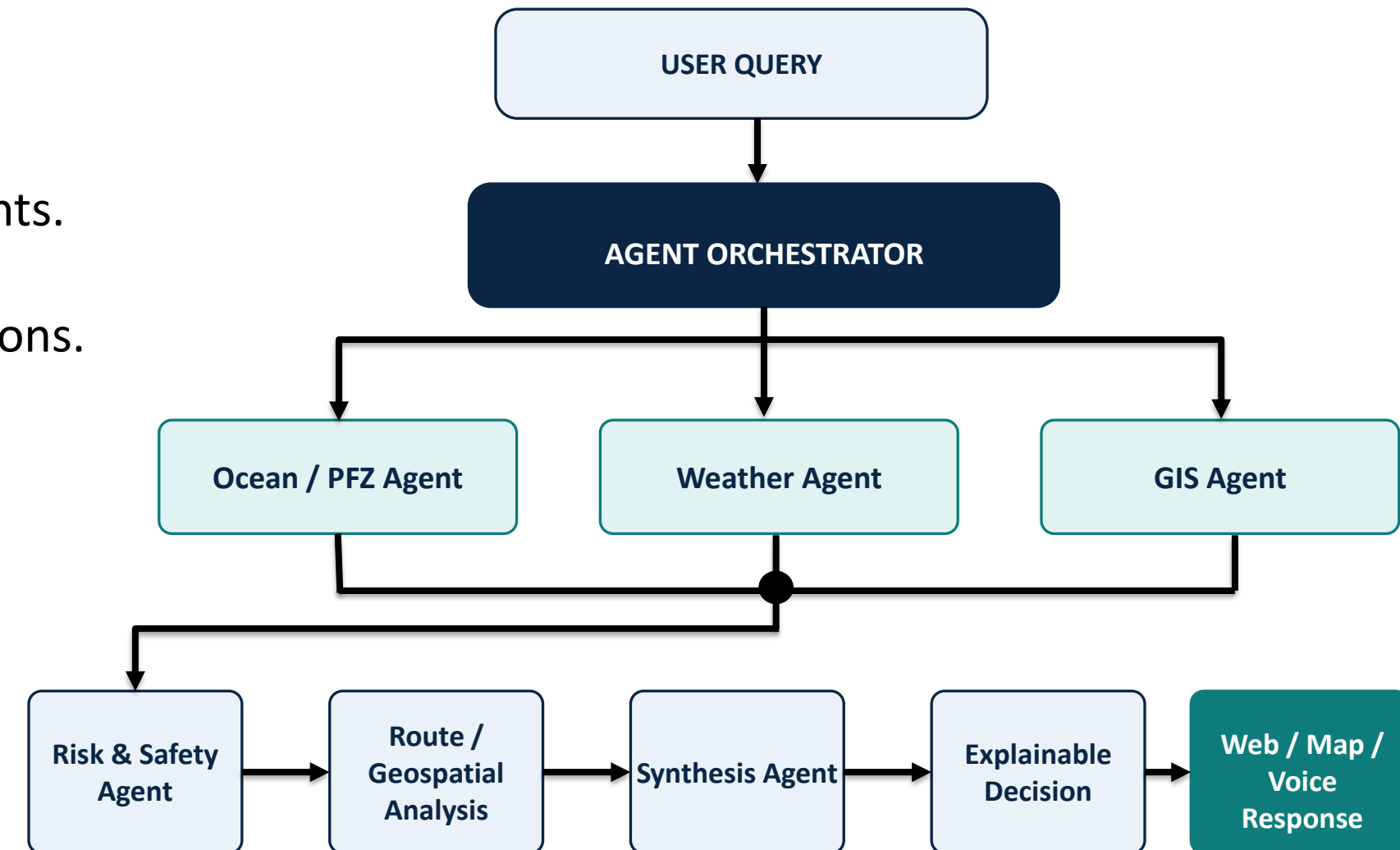
- **One Unified Platform** – All features are available in one place.
- **AI-Based Analysis** – Automatically analyses data and extracts insights.
- **Interactive Dashboard** – Displays results clearly and visually.
- **Smart Decision Support** – Helps users make faster, informed decisions.

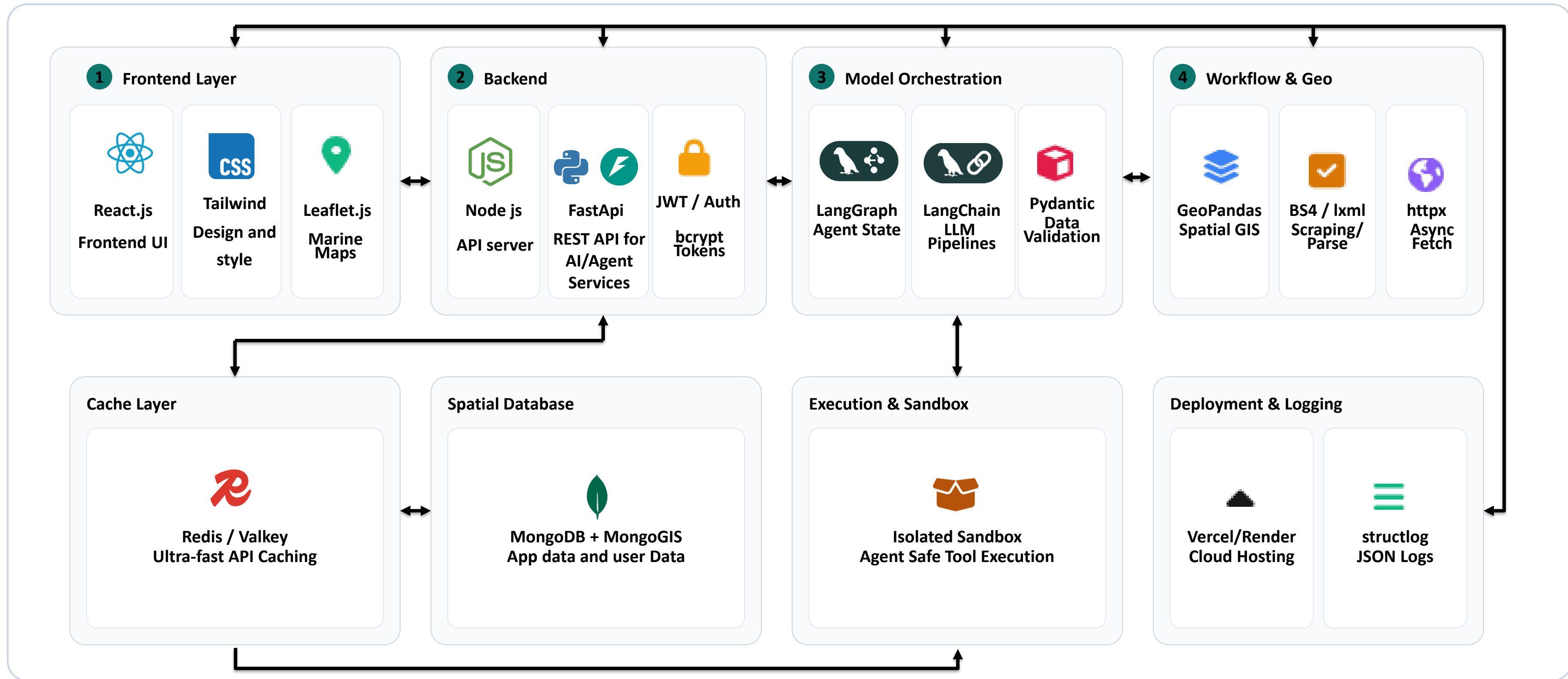
2 How It Addresses the Problem

- **Centralizes Information** – Brings all data into one platform.
- **Reduces Manual Work** – Automates data processing and analysis.
- **Improves Accuracy** – Uses AI to identify important information.
- **Enables Quick Action** – Provides clear insights for faster decisions.

3 Innovation & Uniqueness

- | | | | |
|---|--|---|--|
| <ul style="list-style-type: none">• One-Platform Approach
Integrates all major features and services into a single platform. | <ul style="list-style-type: none">• AI-Powered Automation
Uses AI to automate data analysis and reduce manual effort. | <ul style="list-style-type: none">• Real-Time Insights
Provides quick and meaningful insights from the available data. | <ul style="list-style-type: none">• Multi-Agent AI System
Multiple specialized AI agents work together to analyses data, solve tasks, and provide accurate results. |
|---|--|---|--|





Flow: ① User Query → ② Planner Agent → ③ Parallel Specialist Agents → Data Validation ↔ Risk/Safety Agent → ④ Geo/Route Agent → Synthesis Agent → Explainable Answer.

AGENTS : Ocean/PFZ Agent, Weather Agent, GIS/Geo Agent, Risk & Safety Agent, Route Agent, Orchestrator/Synthesis Agent.

Feasibility

- **INCOIS** already provides PFZ, SST, chlorophyll, and ocean-state advisory datasets.
- **Modular agents** independently fetch weather, PFZ, geospatial, and safety information.
- **React, RESTAPI, Python, Redis, and PostGIS** enable rapid MVP development.
- **Deterministic safety rules** prevent LLM-generated recommendations from becoming unsafe.

Viability

- **ORCA** turns complex marine datasets into simple, actionable go/no-go decisions.
- **Valkey** caching reduces API calls, latency, infrastructure cost, and repeated data fetching.
- **The same platform scales** across coasts, languages, agencies, and stakeholder roles.
- **Source timestamps and risk** explanations improve trust, accountability, and adoption.

Use Cases

- **Safe fishing trip planning:** Receives go/no-go verdict, risk factors, and suggested zone.
- **Hazard-aware route guidance:** Vessel operator gets weather-safe, hazard-aware route to Potential Fishing Zone.
- **Coastal authority queries** cyclone impact; receives affected areas and alerts.
- **Boat near maritime boundary** gets automatic geofence warning and course correction.

Supporting Facts

- **Marine, weather, satellite, and GIS** data currently exist across multiple portals.
- **India** faces frequent **cyclones, cloudbursts, and rough-sea** events affecting coastal safety.
- **PFZ advisories** remain **underused** due to fragmented weather and safety information.
- **Multi-agent AI** enables autonomous planning, reasoning, and explainable marine decisions.



Impacts

- **Improves fishing safety** through timely weather, wave, and hazard intelligence.
- **Increases catch efficiency** using integrated Potential Fishing Zone recommendations.
- **Strengthens coastal disaster preparedness** through localized risk alerts and mapping.
- **Reduces maritime boundary violations** using real-time geofencing and route guidance.



Benefits

- **Enables faster, data-driven decisions** for fishers and coastal authorities.
- **Combines fragmented marine datasets** within one accessible decision-support platform.
- **Delivers multilingual, explainable recommendations** through chat and voice interfaces.
- **Saves fuel, time, and operational costs** through safer route optimization.



Challenges

- **Integrating heterogeneous satellite**, weather, GIS, and oceanographic data streams.
- **Maintaining real-time accuracy** during rapidly changing marine weather conditions.
- **Providing reliable connectivity** in remote offshore fishing locations.
- **Building user trust** in AI-generated safety and navigation recommendations.



Solutions

- ✓ **Use APIs and standardized formats** for seamless multi-source data integration.
- ✓ Apply **continuous forecast updates**, validation, and confidence-based risk scoring.
- ✓ Enable **offline caching, SMS alerts**, and low-bandwidth communication channels.
- ✓ Provide **transparent explanations**, verified data sources, and human override options.

Research References

- **A Prospectus on Generative Artificial Intelligence in Marine Ecosystem Modelling**
<https://onlinelibrary.wiley.com/doi/full/10.1111/faf.70037>
- **Leveraging artificial intelligence (AI) techniques for sustainable marine resources**
<https://link.springer.com/article/10.1186/s44315-026-00054-0>
- **Generative artificial intelligence and marine ecological monitoring**
<https://www.vliz.be/imisdocs/publications/418950.pdf>
- **A Comprehensive Examination of the Current AI Applications in Ocean Sciences**
<https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2026.1890995/full>
- **From climate data to regulatory decisions: integrating climate AI into marine EIAs**
<https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2026.1870082/full>
- **Satellite-Based Fishing Fleet Tracking: A New Foundation for Global Fisheries Science**
<https://onlinelibrary.wiley.com/doi/10.1111/faf.70103>

Technical References

- **React Documentation** - [React Docs](#)
- **Node.js Documentation** - [Node.js Docs](#)
- **Python Documentation** - [Python Docs](#)
- **LangGraph Documentation** – <https://langchain-ai.github.io/langgraph/>
- **LangChain Documentation** – <https://python.langchain.com/docs/>
- **GeoPandas Documentation** – <https://geopandas.org/>
- **Leaflet Documentation** – <https://leafletjs.com/>
- **FastAPI Documentation** – <https://fastapi.tiangolo.com/>
- **INCOIS Ocean Information Services** – <https://incois.gov.in/>
- **MOSDAC Satellite Data Portal** – <https://mosdac.gov.in/>
- **ISRO Bhuvan Geoportal** – <https://bhuvan.nrsc.gov.in/>

Prototype Link: <https://orca-frontend-51u4.onrender.com/>
GitHub: <https://github.com/avinash2005bharti/NEERDRISTHI>

COMPARISION TABLE

Parameter	Current System	Proposed System — ORCA
Data availability	Fragmented portals	Unified platform
Data format	Raw, technical	Simplified, AI-summarized
Interaction	Dashboards, maps	Chat, voice, natural language
Decision support	Manual interpretation	AI-driven recommendations
Weather input	Separate forecasts	Integrated weather + impact analysis
Ocean/ecology data	Isolated datasets	Correlated multi-source analysis
PFZ usage	Standalone advisories	PFZ + safety + route combined
Navigation	Experience-based	AI-suggested safe routes
Visualization	Multiple maps	Single interactive GIS map
Architecture	Static retrieval	Multi-agent collaborative AI